

Skills Summary

Accumulation Functions and the Fundamental Theorem

Use this reference while completing your worksheet or when studying.

Skill 1 — The FTC Derivative Rule

Rule. If $G(x) = \int_a^x f(t) dt$ with a constant, then

$$G'(x) = f(x).$$

Differentiating undoes the accumulation: you never need an antiderivative. Read the units the same way — if f is litres per minute, G is litres and G' is back to litres per minute.

Example: $G(x) = \int_1^x (t^2 + 3) dt$. Find $G'(4)$.

Step 1. The lower limit is constant and the upper limit is plain x , so the theorem applies directly and no antiderivative is needed.

Step 2. $G'(x) = x^2 + 3$, so $G'(4) = 16 + 3$.

$$\Rightarrow G'(4) = 19$$

Try it: $G(x) = \int_2^x (2t + 5) dt$. Find $G'(3)$.

check: $G'(3) = 11$

Skill 2 — A Variable Limit Needs the Chain Rule

Rule. If the upper limit is a function $u(x)$, its derivative rides along:

$$\frac{d}{dx} \int_a^{u(x)} f(t) dt = f(u(x)) \cdot u'(x).$$

A variable in the *lower* limit flips the sign, because reversing the limits negates the integral.

Example: $H(x) = \int_0^{x^3} \sqrt{t} dt$. Find $H'(4)$.

Step 1. The upper limit is a function of x rather than x itself, so the theorem supplies the integrand and the chain rule supplies the extra factor.

Step 2. $H'(x) = \sqrt{x^3} \cdot 3x^2$, so $H'(4) = 8 \cdot 48$.

$$\Rightarrow H'(4) = 384$$

Try it: $K(x) = \int_0^{x^2} (t + 1) dt$. Find $K'(3)$.

check: $K'(3) = 9$

Skill 3 — Evaluating an Accumulated Amount

Rule. To find how much accumulated — rather than how fast it is accumulating — you *do* need an antiderivative:

$$\int_a^b f(t) dt = F(b) - F(a), \quad F' = f.$$

A negative rate subtracts from the total, so a net accumulation can fall even while the integral is still being taken.

Example: Find $\int_0^4 (2t + 1) dt$.

Step 1. This asks for the total accumulated, not a rate, so find an antiderivative and subtract its values at the two limits.

Step 2. $F(t) = t^2 + t$, so the integral is $(16 + 4) - (0 + 0)$.

$\Rightarrow$ **20**

Try it: Find $\int_0^3 (4t + 2) dt$.

check: 24